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ABSTRACT

The key challenge of image splicing detection is how to identify forged regions by capturing subtle traces from high-quality discriminative. In this paper, we propose two feature extraction branches for image forgery localization, and it is named a Multi-Task Edge-Enhanced Network (ME-Net). Firstly, two feature extraction branches are elaborately designed, taking advantage of the discriminative stacked ConvNeXt and ResNet layers, for both RGB and noise domain features. The two feature extraction branches have different architectures and extract edge features in the RGB domain. Secondly, a Fusion Module of Pyramid Split Double Attention (PSDA) is proposed to effectively fuse hierarchical features from two different domains. Thirdly, an Edge Enhancement Position Attention (EEPA) fusion module is proposed to integrate edge context information into the decoding process, thereby enhancing edge details. Finally, an edge feature fusion Pyramid decoder is constructed for feature reconstruction to generate the predicted mask. Extensive experimental results on CASIA v1.0, CASIA v2.0, Columbia, and NIST2016 demonstrate that the proposed method can accurately localize tampered regions in pixel levels and outperform state-of-the-art methods.

CCS CONCEPTS

• **Computing methodologies**; • **Artificial intelligence**; • **Computer vision**; • **Computer vision problems**; • **Object detection**;

KEYWORDS

Image forgery localization, Two-branch, ConvNeXt, Hierarchical-feature fusion

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1 INTRODUCTION

In recent years, the rise of image editing software has made image editing very easy. After editing, the processing traces are not obvious, and the image content is very realistic, making it difficult for the naked eye to distinguish the authenticity of the image [1–3]. Suppose illegal elements maliciously tamper with the original image and spread the tampered image through the Internet to deceive the victim's trust and create rumors[4]. In that case, it will not only distort the facts but also undermine the stability of network security. Therefore, the proposal of forensics against tampered images has important research significance and application value, which can provide technical support for identifying image integrity and authenticity.

Detection algorithms for image manipulation aim to identify manipulated regions within images, encompassing copy-move, splice, and removal as the main types of image tampering. Copy-move involves duplicating a portion of an image and pasting it elsewhere within the same image. Splice entails copying and pasting a part of one image into another. Removal operations use algorithms to erase an object within the image and restore its background region. Current detection methods fall into two broad categories: those based on traditional features, modeling image acquisition signal processing (e.g., PRNU [5], CFA [6]), and those based on deep learning. Traditional feature-based methods require manual design of detection features and exhibit certain limitations. In contrast, deep learning-based methods leverage convolutional neural networks such as RRU-Net [7] and ManTra-Net [8] for adaptive feature learning and pixel-level tampered region localization, enhancing detection efficiency. However, in convolutional neural networks, the pooling process involves irreversible information filtering, potentially compromising the accuracy of splicing edge features [9].

Various methods aim to blend color, resampling, and noise in the network to capture tampering traces comprehensively. However, current fusion methods, like additive fusion, directly overlay or stack feature images pixel-wise. In practical forensic scenarios, different feature types hold varying importance for tampering localization. Thus, fusion methods should fully leverage the complementarity between these features. To precisely locate tampering regions, detection methods need to grasp the image's detailed structure and accurately identify edge features. This requires a network

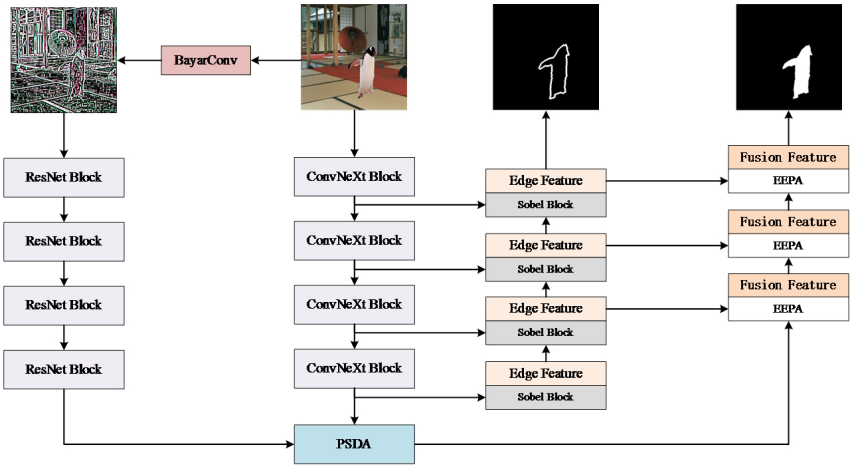

Figure 1: Overview of the proposed ME-Net. ME-Net consists of two feature extraction branches, a PSDA module, an EEPA module, and an edge decoder.

that aggregates multi-level deep features. Previous approaches mainly used binary cross-entropy loss for model training. Yet, when the tampered region occupies a small proportion of the image, the loss function is skewed towards non-tampered regions, resulting in low consistency at boundary pixels and blurry boundaries.

To overcome the abovementioned limitations, we propose a tampering detection network named ME-Net, as in Figure 1. summarized in three-fold: (1) The two-branch edge-aware detection network is proposed, where the RGB branch integrates edge cues into the image segmentation-based forgery localization network through concatenation without pooling operations. (2) The Pyramid Split Double Attention is proposed to achieve more effective fusion of features from different domains of tampered regions and effectively extract finer-grained multiscale information. (3) A method based on edge-guided contextual feature concatenation fusion is proposed to improve the integrity of tampered region detection and reduce interference from non-tampered semantic information. A hybrid loss function is also employed to enhance the network convergence.

2 RELATED WORK

Image forgery detection and localization are distinguished between manipulated images and original images through binary classification at both the image level and the pixel level. Different types of forgeries leave distinct forgery traces in images, such as compression noise, Photo Response Non-Uniformity (PRNU), sensor information [5, 10], and camera model information.

Detection methods often rely on Convolutional Neural Networks (CNNs) for feature extraction, supplemented by auxiliary algorithms for detection and localization. RRU-Net introduces a circular residual structure inspired by human brain mechanisms to enhance CNN learning efficiency. It is worth noting that the Spatial Rich Model (SRM) filter [9] effectively extracts hidden noise in images, demonstrating the potential of utilizing noise features for forgery identification. Similarly, ManTra-Net [8] incorporates BayarConv [10] and SRM to compute features of different image noises, while SPAN [13] improves feature correlation using local self-attention. However, the local attention mechanism limits global context and multi-scale feature utilization. Additionally, a dual-stream Faster R-CNN called RGB-N [11] by Zhou et al. utilizes RGB and noise images for feature fusion through bilinear pooling. MVSS-Net [14] employs trainable dual attention mechanisms for feature fusion and integrates region segmentation and edge detection tasks. PSCC-Net[15] by Liu et al. addresses these issues using channel correlation and a densely connected pyramid structure. Furthermore, CAT-Net[16] combines RGB and DCT-transform features, specifically for JPEG format images. Existing multi-task methods primarily focus on image-level differences, lacking an overall perception of forgery regions. This paper proposes a novel approach to detect and localize image splicing, simultaneously identifying tampered regions and boundary areas, overcoming limitations of previous works.

3 PROPOSED METHOD

3.1 Network Structure

ME-Net utilizes potential forgery clues in different image domains and designs two feature extraction branches to separately extract discriminative features from the RGB and noise domains. The RGB branch, which uses ConvNeXt-B as the backbone network, is focused on extracting features from the subtle boundary artifacts around the tampered regions and performing edge decoding. The noise branch, based on ResNet-50 as the backbone network, captures the noise inconsistency between the tampered and authentic regions. Then, the PSDA module is used to fuse the concatenated features of the two domains. Finally, the tampered regions are obtained through the edge pyramid decoding of the fused features.

3.2 Edge and Noise-Aware Structure

To enhance the integrity of the network in locating tampered regions, ME-Net utilizes an edge-supervised network as non-semantic contextual information to assist the primary network decoding and

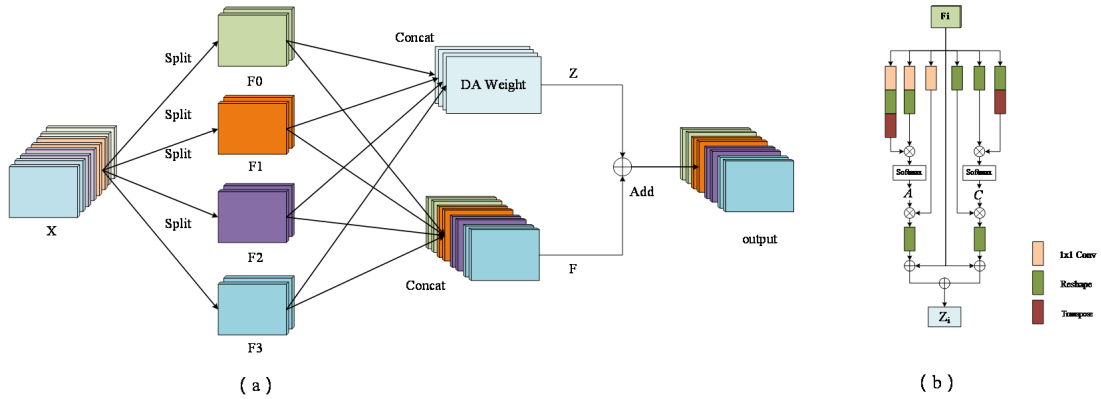

Figure 2: Pyramid Split Double Attention (PSDA)

can also extract tampered region edge masks. For an input image, ConvNeXt-B is used to extract tampering region features from RGB images and concatenate edge features. ConvNeXt is a fully convolutional network that can avoid image information extraction loss caused by pooling and effectively extract edge information. Each layer's output structure calculates corresponding edge features using the Sobel operator module. The basic idea of the Sobel layer is to distinguish edge-related pixels from other pixels in a given feature map by using edge-related weights. The feature map through a classic Sobel filter obtains a Sobel feature map, which is widely used to identify candidate edge pixels[17]. Finally, the computed edge features are decoded layer by layer using a pyramid decoding approach, resulting in edge features for each layer.

In splicing, the noise in different images is often different, and therefore, the rough range of the splicing tampering area can be extracted by the noise. This paper uses BayarConv to convert the RGB domain to the noise domain and uses ResNet-50 as the backbone for encoding. ResNet extracts noise networks, which can reduce computational complexity and highlight noise texture information due to the inclusion of pooling layers. According to Bayar and Stamm [12], BayarConv is designed to enhance the inconsistency in noise between the manipulated area and the real area in a given image. To this end, the BayarConv layer is implemented as a set of trainable prediction error filters. The response of each filter is designed as the error between the center pixel value of the filter window and the linear combination of the remaining pixel values in the window.

3.3 Branch Fusion by Pyramid Split Double Attention

To enhance the fusion of features from diverse domains, we propose a novel algorithm that leverages the DA attention mechanism and incorporates the multi-scale concept. We present a feature fusion enhancement module, Pyramid Split Double Attention (PSDA), which utilizes a trainable pyramid dual attention mechanism for the effective fusion of double-stream tampered area features. This approach facilitates the extraction of fine-grained multi-scale information and fosters the establishment of longer-distance channels and position dependency relationships. The PSDA module adaptively accentuates significant feature information while suppressing irrelevant details.

The DA weight module has two attention mechanisms working in parallel: position attention (PA) and channel attention (CA), see Figure 2(b). PA adaptively refines feature information at each position through a weighted amalgamation of features across various positions.[18] Concurrently, CA deftly correlates channel-wise features to selectively enhance interdependent channel maps.

As illustrated by Figure 2, the PSDA module is mainly implemented in three steps. First, the input feature map is segmented by channels, following which multi-scale feature extraction occurs for the spatial data within each channel feature map. Second, the DA weight module extracts channel and positional fusion attention for different scale feature maps, resulting in fusion vectors for channel and positional attention on different scales, and finally, the recalibrated weights and corresponding features. Maps are elementwise added to output a multi-scale feature map weighted by attention. The model uses a multi-branch approach to extract spatial information from the input feature map [19], allowing for more comprehensive location information. Each branch has an input channel dimension of C. Processing is carried out in parallel across multiple scales, obtaining feature maps containing a single kernel type. Different spatial resolutions and depths are generated using a multi-scale convolution kernel in a pyramid structure.

Input feature map $X \in R^{C\times H\times W}$, slice feature map X into S parts, and $X_0, X_1, \ldots, X_{S-1}$ to represent each segmented feature map. The number of channels in each segmented section is $C_s = C/S$ and the feature map $X_i \in R^{C_S\times H\times W}$, $i = 0, 1, \ldots, S-1$. Note that C should be divisible by S. For each channel feature map divided, multi-scale convolution is applied to extract feature information from different scale feature maps, and multi-scale convolution kernels are used to group convolutions to reduce parameter size. At the same time, adaptively select groups based on the size of the convolutional kernel. the multi-scale feature map generation function is given by

$$F_i = Conv(K_i \times K_i, G_i)X, i = 0, 1, 2, \ldots, S-1 \quad (1)$$

Where the kernel size $K_i = 2\times(i+1)+1$, group size $G_i = 2^{\frac{K_i-1}{2}}$ and the feature map with different scales $F_i \in R^{C_i\times H\times W}$. By extracting the position and channel attention weight information from

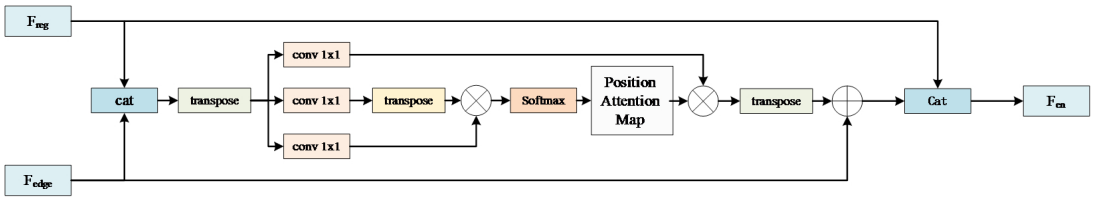

Figure 3: Edge Enhancement Position Attention (EEPA)

the multi-scale processed feature map, the attention weight vectors for different scales are obtained. Mathematically, the attention weight vectors can be represented as

$Z_i = DAWeight(F_i),\ i = 0, 1, 2, \ldots, S-1$ (2)

Where $Z_i = R^{C_i \times H \times W}$ is the attention weight, the DAWeight module calculates the attention weight from the input feature map at various scales. Subsequently, the multiple-scale feature maps are added to the attention vectors that undergo weight calibration, enhancing long-term channel and spatial attention dependencies and facilitating information interaction between multi-scale channel and spatial attention. The weighted feature maps as

$Y_i = F_i \oplus Z_i,\ i = 0, 1, 2 \ldots S-1$ (3)

Where Y_i represents the feature map representing the obtained multiscale double attention weights, $\oplus$ denotes element-wise addition. Finally, the feature maps, after weighting the multiscale channel and position attention, are concatenated along the dimension, and the resulting feature map is outputted, which provides richer multiscale information. The process of obtaining the refined output can be written as

$Out = Cat([Y_0, Y_1, \ldots, Y_{S-1}])$ (4)

As demonstrated by the above analysis, the PSDA module can integrate multiscale spatial and channel attention information into each feature group block through cross-channel integration. Consequently, the PSDA module can achieve better information interaction between global and local attention.

3.4 Attention Edge Feature Fusion Enhancement and Loss

For different types of features, the commonly used fusion methods are cascading or adding, but this fusion method may not be the best. To this end, the Edge Enhancement Position Attention (EEPA) fusion module for edge contextual feature enhancement is proposed. Due to the significant importance of stitching edge features F_{edge} in locating tampered areas, the tampered area features F_{reg} and F_{edge} are first concatenated. The position dependency is calculated through the position attention module, and the spatial dependency between any two positions of the feature map is captured through a self-attention mechanism. The weighted sum is used to aggregate and update the features of all positions, where the weight is determined by the similarity of the corresponding two positions' features. Finally, F_{reg} and F_{edge} are superimposed at the corresponding positions to highlight edge information on the tampered area features. Establish rich non-semantic contextual relationships on local features, encode broader contextual information into local features, enhance representation ability, and concatenate them to obtain the enhanced feature $F_{en} \in R^{C \times H \times W}$ as

$F_{en} = \{[PA]_{reg+edge} \oplus F_{edge}\ ;\ F_{reg}\}$ (5)

Where $[PA]_{reg+edge}$ stands for Position Attention, the general framework of our EEPA module is shown in Figure 3.

The loss function proposed by the algorithm consists of two parts: forgery loss L_{reg} and edge loss L_{edge}, and both of them are defined by Hybrid Loss. The ET is trained by minimizing the total loss L_{total}, where $L_{total} = L_{reg} + L_{edge}$. The hybrid loss function refers to the combination of Binary Cross Entropy (BCE)[20], Structural Similarity (SSIM)[21], and Intersection-over-Union (IoU) losses[22], which allows the model to focus on the salient information of the image at three different levels: pixel-level, patch-level, and map-level.

4 EXPERIMENTAL RESULTS

4.1 Experimental Settings

To comprehensively assess the performance in image stitching localization, four public datasets were adopted in the experiment, namely, CASIA v1.0, CASIA v2.0 [23], Columbia [24], and NIST2016 [24]. Specifically, CASIA v1.0 and CASIA v2.0 both contain splicing and copy-move images. There are a total of 5123 compressed images in CASIA v2.0. CASIA v1.0 has 921 images, with the splicing portion of CASIA v1.0 images being organized as the CASIA v1.0 Splicing dataset, consisting of 461 images. In addition, a CASIA v2.0 enhanced dataset containing 91,000 images was generated as the training dataset, with amplified tampered region features [26], while a portion of CASIA v1.0 and CASIA v1.0 Splicing was used as the testing dataset. The CASIA v2.0 enhanced dataset uses all forged regions as candidate donors, selects the most suitable region for each CASIA v2.0 image, and inserts it into the most concealed position. To enlarge and copy the moving image in CASIA v2.0, find the real image of the forged area, then further find the corresponding copy-move image synthesized from the real image and insert the forged area again in the most hidden position. Each image of enlarged CASIA v2.0 contains multiple forged regions. These features make the expanded dataset more adaptable to complex forgery scenarios in practical applications [27]. Columbia [24] is a splicing dataset consisting of 180 images, with 160 images used for training and 20 images used for testing. NIST16 [25] consists of 564 images, including splicing, copy-move, and removal images, with 394 images used for training and 142 images used for testing.

The evaluation of the model was conducted with F1-score, IoU, and AUC as the assessment metrics. During the binarization of the predicted mask, 0.5 was chosen as the threshold for all images.

Table 1: Ablation study on CASIA2.0 enhancement dataset

Variants	Precision	Recall	F1	IOU	AUC
RGB	0.689	0.600	0.611	0.518	0.796
RGB+Edge	0.731	0.652	0.654	0.564	0.841
RGB+Noise+Edge	0.749	0.657	0.662	0.575	0.856
RGB+Noise+PSDA+Edge	**0.750**	0.680	0.678	0.589	**0.856**
RGB+Noise+PSDA+Edge+EEPA	0.736	**0.704**	**0.687**	**0.603**	0.849

Table 2: Comparative study on feature extraction networks

RGB	Noise	CASIA v1.0	CASIA v1.0-splicing
ResNet	ConvNext	0.503/0.730	0.735/0.888
ResNet	ResNet	0.518/0.758	0.740/0.861
ConvNext	ConvNext	0.670/0.826	0.846/0.931
ConvNext	ResNet	**0.689/0.855**	**0.852/0.941**

Table 3: Performance on pixel-level manipulation detection

Method	CASIA v1.0	CASIA v1.0-splicing	Columbia	NIST16
RGB-N[11]	0.408/0.795	-/-	0.697/0.851	0.722/0.937
RRU-Net[7]	0.405/0.708	0.615/0.825	0.873/0.915	-/-
SPAN[13]	0.382/0.838	-/-	-/0.936	0.582/0.961
PSCC-Net[15]	0.408/0.795	-/-	-/0.982	0.742/**0.991**
ManTra-Net[8]	-/0.817	-/-	-/0.824	-/0.795
CAT-Net[16]	0.380/0.647	0.604/0.813	0.915/0.965	-/0.930
MVSS-Net[14]	0.477/0.730	0.701/0.867	0.950/0.965	0.799/0.939
ME-Net	**0.689**/**0.855**	**0.853**/**0.942**	**0.964**/**0.985**	**0.905**/0.975

All the input images are resized to 256×256 and trained on an NVIDIA RTX3080 GPU. The Multi-Task Edge-Enhanced for image splicing localization is implemented with the Pytorch framework[28]. The RGB and Noise feature extractors are initialized using ConvNeXt-B[29] and ResNet-50, respectively. SGD optimizer with the learning rate adjusted by the polynomial decay strategy $lr = lr_0(1 - Iter_{current}/Iter_{total})^{0.9}$, where $Iter_{current}$ denotes the current number of iterations, $Iter_{total}$ denotes the total number of iterations, and the initial learning rate $lr_0 = 0.001$.

4.2 Ablation Study and Robustness Analysis

The experiment validated the main module's effectiveness using an extended dataset from CASIA2.0 for training and conducted experiments on the CASIA v1.0 dataset. Results in Table 1 show different variants' performance. "RGB" refers to feeding only the last layer's output feature from the RGB branch into the decoder, while "Edge" denotes the output of the edge pyramid decoder from the RGB branch. Ablation experiments demonstrated that adding edge decoding significantly improved all metrics, and further fusion of noise and PSDA features led to additional improvements. Although the addition of EEPA slightly decreased accuracy and AUC, it resulted in a further improvement in the F1 metric by enhancing recall at the expense of precision. The results suggest that the two-branch architecture, Edge decoder, PSDA, and EEPA module all contribute to performance enhancement.

Dual-branch feature extraction employs different feature extraction models. Table 2 reports the RGB branch uses ConvNeXt as the backbone network, while the noise branch uses ResNet-50. This approach not only yields the best results but also reduces computational complexity. It indicates that the fully convolutional ConvNeXt is capable of effectively extracting RGB edge features while preserving more details. Extracting noise using ResNet enables fast identification of inconsistencies in the noise, which leads to better localization of tampered regions.

4.3 Comparison with State-of-the-art Methods

The performance comparison of different tampering localization algorithms on different datasets is in Table IV. "-" indicates not applicable. The performance metric is F1 and AUC. Table 3 reports the compared results on four publicly available datasets, and Figure 4 displays the predicted masks of methods with publicly available code. On the NIST16 dataset, RGB-N, PSCC-Net, and SPAN were fine-tuned, following the same training and testing splits to ensure a fair comparison in this study. Fine-tuning was performed on the CASIAV v2.0 dataset, and testing was conducted on CASIA v1.0, and reference to their respective original papers. Table 3 demonstrates that ME-Net achieved the best performance on each dataset, and

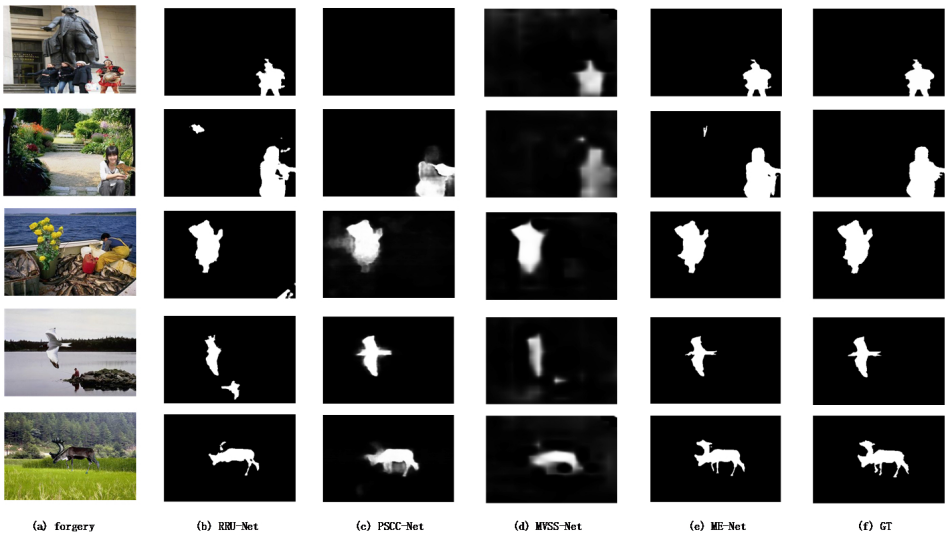

Figure 4: proposed method and other comparative methods

Figure 4 further reveals that ME-Net provides more comprehensive boundary localization of forged regions.

5 CONCLUSION

In this work, a new ME-Net was proposed to tackle the challenges of image-splicing tampering techniques. It adopts a dual-stream feature extraction approach and utilizes non-semantic contextual information from image edges for decoding. Each edge feature serves as prior information to assist in predicting the mask of the next scale. Additionally, the PSDA attention design is employed to spatially and channel-wise fuse the features extracted from two domains, providing holistic feature information. Moreover, the EEPA adaptive fusion of edge features assists in decoding, making the model more suitable for tampering detection and localization. Extensive experiments demonstrate that ME-Net outperforms state-of-the-art methods in both detection and localization. For future work, the detection and localization of various tampering techniques will be implemented, and strategies to estimate the uncertainty of predicted operation masks will be developed further to improve the generalization performance of tampering detection and localization.

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